

Econ 8220: Problem Set #3

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1 Introduction

The main objective of this problem set is to get a feel for the estimation and identification in models where the parameters are partially identified. Specifically, we study Amazon's network of fulfillment centers (FCs) and how the network affects taxes, revenues, costs, and ultimately Amazon's optimal network choices. This is in the spirit of Houde et al. (2023).

Complete this problem set by answering the questions below. When you are finished, submit your completed work as a pdf (or multiple pdfs) with naming convention '**PS3_[insert last name]**'. For example, 'PS3_Newberry.pdf'. Feel free to have one pdf for your write-up and one for your code with the labels "_code" and "_writeup", respectively. Feel free to work together but turn in your own write-up. You can use any programming language you wish.

Part of the purpose of these problem sets is to help you learn how to write up an academic paper (especially an empirical one), so please include derivations of estimating equations, details of how you estimate the model, intuition about the answers, etc., in your write-up.

2 Set Up and Data Description

Each market (county) has a representative consumer i , who chooses how much to spend in year t at four different purchasing options (called "modes"): Amazon.com ($j = 1$), taxed online competitors ($j = 2$), untaxed online competitors ($j = 3$), and offline competitors ($j = 0$). Demand for each mode j is a function of consumer preferences for that mode (α_{ijt}) and the effective tax rate that the consumer pays on purchases from mode j in year t (τ_{ijt}).

The estimating equation has the log of the ratio of spending on Amazon to the outside good as the dependent variable, so the predicted log ratio is given by:

$$\log \frac{e_{ijt}}{e_{i0t}} = \alpha_{ijt} + (1 - \sigma) (\log(1 + \tau_{ijt}) - \log(1 + \tau_{i0t}))$$

The model then predicts the ratio of spending in levels as:

$$\frac{e_{ijt}(a_t)}{e_{i0t}(a_t)} = \exp(\alpha_{ijt} + (1 - \sigma) (\log(1 + \tau_{ijt}) - \log(1 + \tau_{i0t})))$$

where a_t represents Amazon's network of fulfillment centers in year t . The network varies the effective taxes that each consumer pays. Specifically, if Amazon has an FC in consumer i 's state in year t , then they pay the sales tax rate in their county/year on purchases from Amazon. If Amazon does not have an FC in their state in year t , they do not pay sales tax. For $j = 2$ and $j = 0$, consumers always pay sales tax, and for $j = 3$ consumers never pay sales tax. Assume that $\sigma = 1.5$.

Total revenue for Amazon is:

$$R_t(a_t) = \sum_i M_{it} e_{i1t}(a_t)$$

where M_{it} is the number of households in county i in year t .

Total labor costs are given by:

$$C_t^\ell(a_t) = \sum_i w_{it} \ell_{it}(a_t)$$

where w_{it} is the wage in county i in year t and $\ell_{it}(a_t)$ is the amount of labor hired in that county. The amount of labor hired depends on the network; if Amazon does not have a facility in that county/year, labor is 0.

Total land costs are given by:

$$C_t^k(a_t) = \sum_i r_{it} k_{it}(a_t)$$

where r_{it} is the rent per square foot in county i in year t and $k_{it}(a_t)$ is the total square footage Amazon rents in that county. If Amazon does not have a facility in that county/year, the square footage is 0.

The fixed costs for Amazon consist of a shipping cost (a function of distance) and a cost based on population density:

$$F_t(a_t) = \sum_i \gamma d_{it}(a_t) + \omega \text{PopDens}_{it}(a_t)$$

where d_{it} is the shipping distance between the closest facility and county i , and $\text{PopDens}_{it}(a_t)$ is the effective population density of county i . Both depend on the network; if Amazon does not have a facility in county i in year t , the effective population density is 0.

Amazon chooses the sequence of openings from $t = 0$ (1999) to $t = \infty$, denoted $a = \{a_0, a_1, \dots, a_\infty\}$, to maximize its discounted stream of profit:

$$a^* = \max_a \Pi(a) = \sum_{t=0}^{\infty} \beta^t (\mu R_t(a_t) - C_t^\ell(a_t) - C_t^k(a_t) - F_t(a_t))$$

where $\mu = 0.15$ and $\beta = 0.95$.

The data are separated into several files. The file `data_observed.csv` contains all data associated with the observed network, including everything needed to calculate the discounted sum of (1) variable profit (margin $\times$ revenue $-$ land cost $-$ labor cost), (2) population density, and (3) shipping distance under the observed network. The file `data_observed_dict.csv` defines each variable. The files `perturb_data#.csv` contain data for the counterfactual swaps, separated into 4 files of roughly 500 swaps each. Each swap only contains the counties/years in which the swap resulted in a change in one or more important variables. The file `data_perturb_dict.csv` defines each variable in these files.

1. Import the observed data (`data_observed.csv`) and the perturbed data files (`perturb_data1.csv` through `perturb_data4.csv`) from eLC. Review the data dictionaries to understand each variable.
2. For the observed network and for each perturbation, calculate the discounted sum of: gross revenue, labor costs, land costs, shipping distance, and effective population density. Also calculate the discounted population-weighted average tax rate paid on Amazon purchases. Calculate the pairwise correlations among all of these components. Do they make sense? Provide intuition for each notable correlation.

3 Estimating the Model

Revealed preferences suggest that any perturbation from a^* is suboptimal, so that $\Pi(a^*) \geq \Pi(a_s)$. This implies that, for perturbation s with network a_s :

$$y(a^*, a_s) - \gamma x^d(a^*, a_s) - \omega x^p(a^*, a_s) \geq 0$$

where $y(a^*, a_s)$ is the difference in the discounted sum of variable profit (gross revenue net of land and labor costs), $x^d(a^*, a_s)$ is the discounted sum of shipping distance, and $x^p(a^*, a_s)$ is the discounted sum of effective population density. Assuming classical measurement error in variable profit:

$$\tilde{y}(a^*, a_s) - \gamma x^d(a^*, a_s) - \omega x^p(a^*, a_s) + \epsilon_s \geq 0$$

Since the error is mean zero, we form moment inequality m :

$$M_m(\omega, \gamma) = \frac{1}{S} \sum_s \tilde{y}(a^*, a_s) - \gamma \frac{1}{S} \sum_s x^d(a^*, a_s) - \omega \frac{1}{S} \sum_s x^p(a^*, a_s) \geq 0$$

1. **Isolating γ (20 points).** For the first exercise, let us isolate each parameter to get preliminary estimates.

- (a) Select inequalities where the discounted sum of effective population density does not change much from the observed (e.g., below the 25th percentile of absolute changes). With this restriction, assume $x^p(a^*, a') = 0$ for this set of inequalities, leaving:

$$\frac{1}{S} \sum_s \tilde{y}(a^*, a_s) - \gamma \frac{1}{S} \sum_s x^d(a^*, a_s) \geq 0$$

Calculate the upper and lower bound of γ using these perturbations. Remember that whether you are calculating the upper or lower bound will depend on the sign of $x^d(a^*, a_s)$. You may further refine the groups based on $x^d(a^*, a_s)$ and an additional shifter of $\tilde{y}(a^*, a_s)$ (e.g., the population-weighted tax rate). Provide intuition for your choices.

- (b) Now perform the same exercise without conditioning on population density changes. How do your answers change? Provide intuition.

2. **Simultaneously estimating γ and ω (30 points).** Now estimate bounds on both parameters simultaneously using a grid search.

- (a) Select inequalities that represent the trade-offs identifying each parameter. Use your findings from Exercise 1 to guide selection of inequalities for γ (distance). For ω (population density), think carefully about the benefit of entering a densely populated area (despite higher taxes and density costs) and use this trade-off to select inequalities. You should arrive at four moments (one for each side of the trade-off for each parameter), denoted m .
- (b) For each point on the grid, calculate the objective function:

$$Q(\omega, \gamma) = \sum_m \min(0, M_m(\omega, \gamma))^2$$

Save the grid values of the parameters, the value of $\min(0, M_m(\omega, \gamma))$ for each of the four moments, and the value of Q to a CSV file.

- (c) Report the identified set (i.e., the range of (γ, ω) that minimizes Q). Discuss your findings. Are the bounds well-identified? What might explain any difficulties in estimation?